

MAT111 Assignment 1

Submission link: Forms

Deadline: November 11, 2025

Problem 1. Consider the relation $R = \{(x, x) : x \in \mathbb{Z}\}$ on $\mathbb{Z}$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why. What familiar relation is this?

Problem 2. Consider a relation $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x - y \in \mathbb{Z}\}$ on $\mathbb{R}$. Prove that this relation is reflexive, symmetric and transitive.

Problem 3. Prove or disprove: If a relation is symmetric and transitive, then it is also reflexive.

Problem 4. There are 16 possible different relations R on set $A = \{a, b\}$. Describe all of them.

Problem 5. Define a relation $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : |x - y| \leq 1\}$ on $\mathbb{Z}$. Is R reflexive? Symmetric? Transitive? If a property does not hold, say why. What familiar relation is this?

Problem 6. Define a relation R on $\mathbb{Z}$ as xRy if and only if $x^2 + y^2$ is even. Prove that R is an equivalence relation.

Problem 7. Suppose R is an equivalence relation on a finite set A , and every equivalence has the same cardinality m . Express $|R|$ in terms of m and $|A|$.

Problem 8. Prove or disprove: If R is an equivalence relations on an infinite set A , then R has infinitely many equivalence classes.

Problem 9. Suppose R is an equivalence relation on a set A , with four equivalence classes. How many different equivalence relations S on A are there for which $R \subseteq S$?

Problem 10. Describe the partition of $\mathbb{Z}$ resulting from the equivalence relation $\equiv (\text{mod } 4)$.

Problem 11. Consider the partition $P = \{\{\dots, -4, -2, 0, 2, 4, \dots\}, \{\dots, -5, -3, -1, 1, 3, 5, \dots\}\}$ of $\mathbb{Z}$. Let R be the equivalence relation whose equivalence classes are the two elements of P . What familiar equivalence relation is R ?

Problem 12. Write the addition and multiplication tables for $\mathbb{Z}_6$.

Problem 13. Let $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Show that $a + c \equiv b + d \pmod{n}$.

Problem 14. Let $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Show that $a \cdot c \equiv b \cdot d \pmod{n}$.

Problem 15. Prove that if p is prime then, modulo p , every nonzero number has a unique multiplicative inverse; i.e., if x is not a multiple of p then there is a unique $y \in \{1, 2, 3, \dots, p-1\}$ such that $xy \equiv 1 \pmod{p}$.

Problem 16. Sketch the set $X = [-1, 3] \times [0, 2]$ on plane $\mathbb{R}^2$. On a separate drawing, shade the sets $\overline{X}$ and $\overline{X} \cap ([-2, 4] \times [-1, 3])$.

Problem 17. Write the following Venn diagrams:

- $(A \cap B) \cup (A \cap C)$
- $\overline{A \cap B}$
- $(A \cap B) - C$
- $(A - B) \cup C$

Problem 18. If $J \neq \emptyset$ and $J \subseteq I$, check if the following are true. Justify your answer.

- $\bigcup_{a \in J} A_a \subseteq \bigcup_{a \in I} A_a$
- $\bigcap_{a \in J} A_a \subseteq \bigcap_{a \in I} A_a$
- $\bigcap_{a \in I} A_a \subseteq \bigcap_{a \in J} A_a$

Problem 19. Between $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and $\mathbb{R}$, which sets have a smallest element? What about greatest elements? Which sets can be used for the **well-ordering principle**?

Problem 20. Can you construct a set which has no greatest element but is bounded above? In other words, the set has an upper bound.

Definition: (Upper Bound) A set X has an upper bound if there exists an element x such that any element of X is less or equal to x .

Problem 21. Find the least upper bound of the set $A = \{x \in \mathbb{Q}^+ : x^2 < 2\}$. If we consider all the upper bounds as a set, then the **Least Upper Bound** is the smallest element of the upper bound set.

Problem 22. Sketch the following sets on a x - y plane:

- $\{(x, y) : x, y \in \mathbb{R}, x^2 + y^2 = 1\}$
- $\mathbb{Z} \times \mathbb{Z}$
- $\mathbb{N} \times \mathbb{Z}$
- $\mathbb{R} \times \mathbb{Z}$

Problem 23. Explain why $\mathbb{R} \times \mathbb{N} \subseteq \mathbb{R} \times \mathbb{R}$.

Problem 24. Find the cardinalities of the following sets:

- $\{X \in \mathcal{P}(\{1, 2, 3\}) : 2 \in X\}$ $\{X \in \mathcal{P}(A) : |X| \leq 1\}$ where $|A| = m$
- $\{X \subseteq \mathcal{P}(A) : |X| \leq 1\}$ where $|A| = m$

Problem 25. Is the statement $\mathbb{R} \times \mathbb{Z} \cap \mathbb{Z} \times \mathbb{R} = \mathbb{Z} \times \mathbb{Z}$ true? What about $\mathbb{R} \times \mathbb{Z} \cup \mathbb{Z} \times \mathbb{R}$?

Problem 26. Using the Axioms of Set theory construct:

- Intersection between two sets.
- Subtraction of two sets.
- Complement of a set.

Problem 27. Between the following statements, determine what is the difference between them and state which of them is/are correct.

- $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}, m = n + 5$
- $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, m = n + 5$

Problem 28. Translate each of the following sentences into symbolic logic

- *I don't eat anything that has a face*
- *Everything is funny as long as it is happening to somebody else*
- *If $\sin(x) < 0$, then it is the case that $0 \leq x \leq \pi$.*
- *For every positive number ϵ there is a positive number M for which $|f(x) - b| < \delta$ implies $|f(x) - f(a)| < \epsilon$.*
- *For every even number n , there exists two prime numbers p, q such that $n = p + q$*

Problem 29. Negate the following sentences.

- *I don't eat anything that has a face*
- *Everything is funny as long as it is happening to somebody else*
- *You can fool all of the people all of the time.*
- *For every prime number p , there is another prime number q such that $p < q$.*
- *For every even number n , there exists two prime numbers p, q such that $n = p + q$*
- *If f is a polynomial and its degree is greater than 2, then f' is not a constant.*

Problem 30. Well this is not a problem. In set theory, by cardinality we can find the size of a set. Say $A = \{1, 2, 3\}$ has the cardinality of 3 because it has 3 elements in it. We consider two sets has the same cardinality if we can form a bijection between them. Lets consider two set $\mathbb{N} = \{1, 2, 3, \dots\}$ and $X = \{2, 4, 8, 16, \dots\}$ a set which contains the power of 2. X is clearly a subset of $\mathbb{N}$ here. However, the cardinality of these two sets are equal. How?